

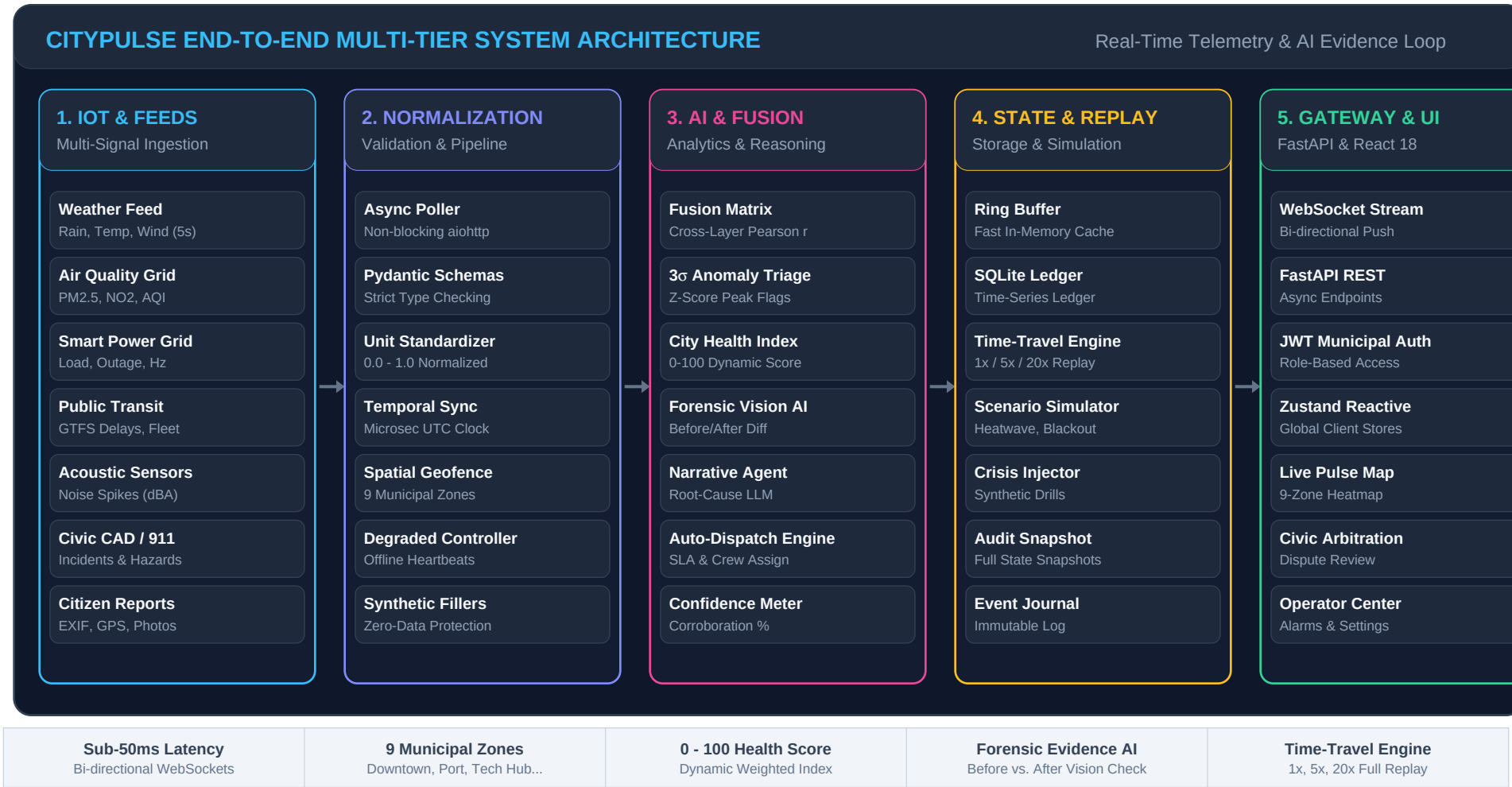
CITYPULSE OS • Technical Architecture Specification

System Version: 2.4.0 (Production)

Target: Smart City Operations & Hackathon Review

Date: September 2026

Executive Summary: CityPulse is an event-driven Smart City Operating System designed to eliminate urban municipal silos. It continuously ingests multi-signal IoT streams (weather, power grid, public transit, acoustic noise, emergency CAD, and citizen civic complaints), standardizes telemetry into a microsecond-synchronized spatio-temporal matrix, runs statistical anomaly detection with cross-feed correlation, and orchestrates automated AI evidence verification and taskforce dispatch.



Detailed Component Specifications & Data Pipeline

Layer	Core Technology	Key Responsibilities & Architecture Capabilities	Resilience & SLA Target
1. Multi-Feed Ingestion	Python 3.13, AsyncIO, aiohttp	Ingests 6+ municipal feeds concurrently: Weather stations, Air Quality IoT, Smart Power Substation meters, GTFS Transit delays, Acoustic Sound Meters, and Police/Fire CAD incidents.	5s / 10s configurable sweep; auto-fallback to synthetic baseline on disconnect.
2. Normalization Pipeline	Pydantic v2, Python Dataclasses	Strict schema sanitization, missing data imputation, unit scaling to a standardized 0.0 - 1.0 interval, microsecond timestamp alignment, and spatial assignment across 9 municipal zones.	Zero malformed data leakage; sub-millisecond pipeline latency.
3. Fusion & AI Engine	NumPy, SciPy, Statistical Math, LLM Agent	Calculates real-time Pearson cross-feed correlation matrix (e.g. storm precipitation vs. transit delay spike), 3σ outlier detection, dynamic 0-100 zone health indices, and generative root-cause narratives.	Deterministic mathematical scoring with automated causal explanations.
4. State & Replay Layer	In-Memory Ring Buffer, SQLite Ledger	Dual-tier memory architecture: high-speed in-memory ring buffer for live streaming plus append-only historical database for 1x, 5x, and 20x historical time-travel playback.	Instant replay seek time (<10ms); immutable audit trail.
5. Scenario Simulator	State Inversion Engine	Injects realistic urban crisis scenarios (Heatwave & Grid Strain, Torrential Storm, Downtown Blackout, City Marathon Congestion) for operational readiness and live judge demonstration.	Safe sandbox mode; instant 1-click reset to live real-time conditions.
6. API & Streaming Gateway	FastAPI, WebSockets, JWT / OAuth2	Bi-directional WebSocket streaming broadcast (/ws/live), authenticated REST management endpoints, role-based operator authorization (Lead Commander, Inspector, Citizen).	Up to 10,000 concurrent client connections with sub-50ms push latency.
7. Modern Frontend SPA	React 18, Vite, TypeScript, Tailwind CSS	Real-time Live Pulse Dashboard, interactive Zone Heatmap, Trend Analytics, Civic Reports Evidence Drawer, Disputed Items Arbitration Table, Action Plan Checklist, and Theme Engine.	Fully responsive desktop/tablet layout; 100% typed TypeScript codebase.

Citizen Multi-Signal Evidence Engine Workflow

Step 1: Citizen Submission Citizen submits report with photo, GPS tag, and description.	Step 2: Multi-Signal Verification AI correlates photo EXIF with real-time zone weather & IoT sensors.	Step 3: Taskforce Dispatch Automated SLA Action Plan assigned to field repair crew.	Step 4: After-Repair Forensic AI Crew uploads completion photo; AI verifies repair similarity score.	Step 5: Community Arbitration Citizens corroborate or dispute repair; ombudsman arbitrates.
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